

Show your work Geometry Mixed Review — 40 Problems

Think first, then compute. Label diagrams, keep exact values (π , radicals) until the end, and include units.

Mini-Review (use these)

On a Line & in the Plane

Segment addition: If B is between A and C , $AB + BC = AC$.

Distance (1D): $|x_2 - x_1|$. Distance (2D): $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

Midpoint: $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$.

Fraction from A to B : $P = A + f(B - A)$.

Ratio $m : n$ ($A \rightarrow B$): $P = A + \frac{m}{m+n}(B - A)$.

Angles & 2D Area/Perimeter

Straight 180° , right 90° ; vertical $\cong$; linear pair $\Rightarrow$ supplementary; bisector splits equally.

Rectangle: $P = 2(\ell + w)$, $A = \ell w$; Triangle: $A = \frac{1}{2}bh$.

Circle: $C = 2\pi r$, $A = \pi r^2$; Square: $P = 4s$, $A = s^2$.

Rigid Motions (about origin)

Reflect in x -axis: $(x, y) \mapsto (x, -y)$; Translate (a, b) : $(x + a, y + b)$.

Rotate 90° CCW: $(-y, x)$; 180° : $(-x, -y)$; 270° : $(y, -x)$.

Preserve lengths, perimeters, areas, angles.

3D Models

Rect prism: $V = \ell wh$, $SA = 2(\ell w + \ell h + wh)$. Cylinder: $V = \pi r^2 h$, $SA = 2\pi r(h + r)$.

Cone: $V = \frac{1}{3}\pi r^2 h$, $SA = \pi r(r + s)$, $s = \sqrt{r^2 + h^2}$. Sphere: $V = \frac{4}{3}\pi r^3$, $SA = 4\pi r^2$.

Right square pyramid: $V = \frac{1}{3}a^2 h$, $SA = a^2 + 2as$, $s = \sqrt{h^2 + (a/2)^2}$.

Measurement (a few items)

If v is rounded to nearest u , true $\in [v - \frac{u}{2}, v + \frac{u}{2})$.

Sig figs: Add/Sub $\rightarrow$ least precise place; Mult/Div $\rightarrow$ fewest sig figs.

A. Line Placement (1D/2D)

1. A bus line is straight with stops A, B, C and B between. If $AB = 1.8$ mi and $BC = 2.4$ mi, find AC .
2. On a number line, $A = -5$, $C = 9$. With B between, $AB = 2x + 1$, $BC = 3x - 5$. Find x and coord(B).
3. $A = -6$, $B = 10$. Point P is $\frac{3}{4}$ of the way from A to B . Find P .

4. $A = 2$, $B = 18$. Find the point that partitions $\overrightarrow{AB}$ in ratio $3 : 5$ (closer to A).
5. In 2D, $A(-3, 5)$, $B(9, -7)$. Find the point P that is $\frac{3}{5}$ of the way from A toward B .
6. Endpoints of a diameter are $E(-4, 2)$ and $F(6, 2)$. Find circle center, radius, C , and A (leave π).

B. Distance, Midpoint, 2D Area/Perimeter

7. Distance between $C(-6, 4)$ and $D(10, -2)$; then midpoint M .
8. Rectangle $ABCD$ has $A(1, 2)$, $B(9, 2)$, $C(9, -4)$, $D(1, -4)$. Find ℓ , w , then P and A .
9. Opposite vertices of a square are $S_1(-2, 3)$ and $S_2(6, 11)$. Find side length, perimeter, and area.
10. $\triangle ABC$ with $A(0, 0)$, $B(8, 0)$, $C(5, 6)$. Find $A_{\triangle}$ and length AC .

C. Angle Basics

11. Complementary angles measure x° and $(x + 18)^\circ$. Find both.
12. Linear pair: $(3x + 15)^\circ$ and $(5x - 3)^\circ$. Find x and both angles.
13. Vertical angles: $(2y + 14)^\circ$ and $(5y - 13)^\circ$. Find y and the angle measure.
14. Ray $\overrightarrow{BD}$ bisects $\angle ABC$. If $\angle ABD = (x + 20)^\circ$ and $\angle DBC = (2x - 10)^\circ$, find x and $\angle ABC$.

D. Rigid Motions

15. Reflect $A(3, 1)$, $B(7, 4)$, $C(1, 5)$ across the x -axis. List A', B', C' . What happens to perimeter and area?
16. Translate $P(-2, 3)$ by $(5, -4)$ then rotate 90° CCW. Find the final image P'' .
17. Now rotate $P(-2, 3)$ by 90° CCW first, then translate by $(5, -4)$. Are results same as previous? Explain.

18. Which rotation (among 90° , 180° , 270° CCW) sends $U(4, -1)$ to $U'(-1, -4)$? Justify.

E. Mixed Coordinate Geometry

19. A circle centered at the origin passes through $Q(12, 5)$. Find r , C , and A (leave π).
20. Triangle PQR with $P(0, 0)$, $Q(6, 0)$, $R(6, 8)$. Verify right angle using distances; then P and A .
21. Points $R(2, 1)$, $S(9, 4)$, $T(5, -2)$. Find all three side lengths and the perimeter (simplify radicals when reasonable).
22. Rectangle with consecutive vertices $A(0, 0)$, $B(8, 0)$, $C(8, 6)$. (a) Find D . (b) Compute area and perimeter.
23. Show angle at Q is right for $P(-1, 2)$, $Q(3, 0)$, $R(7, -2)$ via slopes or distances.

F. 3D Surface Area & Volume

24. Rectangular prism has $V = 864 \text{ cm}^3$ with base $12 \text{ cm} \times 6 \text{ cm}$. Find h and total SA .
25. Open-top box: $\ell = 18 \text{ cm}$, $w = 12 \text{ cm}$, $h = 10 \text{ cm}$. (a) Cardboard area needed? (b) Ribbon length for one loop around sides (mid-height), no overlap.
26. A cylinder of radius $r = 4 \text{ cm}$ and height h has same volume as a cone with $r = 4 \text{ cm}$, $h = 18 \text{ cm}$. Find h for the cylinder. Which has greater *lateral* area? Brief justification.
27. Right circular cone: $r = 6 \text{ cm}$, slant $s = 10 \text{ cm}$. (a) Find h . (b) Total SA . (c) V .
28. Label wrap: A label of height 8 cm wraps once around a cylinder with no gap/overlap. Label area $= 264 \text{ cm}^2$. Find cylinder radius and height.
29. Sphere has $SA = 324\pi \text{ cm}^2$. Find r and V .

30. Right square pyramid: base edge $a = 14$ cm, slant $s = 13$ cm. (a) Total SA . (b) Height h and V .
31. Two cans must hold $\pi r^2 h = 960\pi$ cm³. Can A has $r = 6$; Can B has $r = 8$. Find each h and compare lateral areas $2\pi r h$.
32. A full cone ($r = 5$, $h = 12$ cm) is poured into a cylinder of radius 5 cm. How high does the water rise?
33. Composite: cylinder $r = 6$ cm, $h = 10$ cm topped with a hemisphere of radius 6 cm. (a) Total V . (b) Exterior SA (include bottom; exclude interior seam).

G. Measurement & Sig Figs (light)

34. A measurement recorded as 7.84 m (nearest 0.01 m). Give the interval for the true value.
35. Compute $(12.07 + 0.3 + 0.025)$ using add/sub rule; then $(3.20)(0.054)$ using mult/div rule. Report sig-figged results.

36. If a distance rounded to the nearest meter is reported as 142 m, what is the maximum possible percent error?
37. Density $\rho = \frac{m}{V}$. $m = 12.0$ g (nearest 0.1 g), $V = 4.0$ cm³ (nearest 0.1 cm³). (i) Sig-fig value of ρ . (ii) Give a plausible range using intervals.